

View Reviews

Paper ID

9

Paper Title

Exploring Agentic AI in Anti-Forensics: Simulation of Evasion Tactics in Digital Investigations

Reviewer #1

Questions

1. What is your overall recommendation for this paper? 3: Strong Accept (Top-tier, award-worthy) 2: Accept 1: Weak Accept / Borderline -1: Weak Reject -2: Reject -3: Strong Reject (Serious technical flaws or incorrect)

Accept

2. Please summarize the main contributions of this paper and its significance in your own words.

The main contribution claimed is the empirical demonstration of the TOCTOU "Time Gap" - showing that an autonomous agent can complete an entire anti-forensic operation (timestomping, secure wiping, log sanitisation) in under 300 milliseconds. The authors not just arguing the gap exists theoretically; they're showing a working agent that reliably exploits it end-to-end. The implicit takeaway is that organisations relying solely on periodically polled FIM are vulnerable and should adopt kernel-level monitoring as a complement

3. Novelty and Significance (1-5): 1 is the worst, 5 is the best. "How novel and original are the ideas, methods, or results presented?" "How significant are the contributions to the field?"

4

4. Technical Quality and Correctness (1-5): 1 is the worst, 5 is the best "Is the technical content correct and sound?" "Are the claims well-supported by evidence (theoretical proofs, experiments, etc.)?" "Is the methodology appropriate and rigorously applied?"

3

5. Clarity and Presentation (1-5): 1 is the worst, 5 is the best "Is the paper well-written and clearly organized?" "Is it easy to understand the main points?"

4

6. Comments for the Authors

I think the analysis should be more systematic (inferential statistic tests through clear metrics with multiple trials) than so far being called experimental instead of just empirical.

8. Your confidence for the review

Medium

Reviewer #2

Questions

1. What is your overall recommendation for this paper? 3: Strong Accept (Top-tier, award-worthy) 2: Accept 1: Weak Accept / Borderline -1: Weak Reject -2: Reject -3: Strong Reject (Serious technical flaws or incorrect)

Strong Accept

2. Please summarize the main contributions of this paper and its significance in your own words.

This paper is clear and technically sound. It discusses an important and emerging issue in digital forensics and enterprise security.

3. Novelty and Significance (1-5): 1 is the worst, 5 is the best. "How novel and original are the ideas, methods, or results presented?" "How significant are the contributions to the field?"

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4. Technical Quality and Correctness (1-5): 1 is the worst, 5 is the best "Is the technical content correct and sound?" "Are the claims well-supported by evidence (theoretical proofs, experiments, etc.)?" "Is the methodology appropriate and rigorously applied?"

4

5. Clarity and Presentation (1-5): 1 is the worst, 5 is the best "Is the paper well-written and clearly organized?" "Is it easy to understand the main points?"

3

6. Comments for the Authors

To further strengthen the paper, I encourage the authors to elaborate on the broader architectural and operational implications of their findings for enterprise security systems.

8. Your confidence for the review

high

Reviewer #3

Questions

1. What is your overall recommendation for this paper? 3: Strong Accept (Top-tier, award-worthy) 2: Accept 1: Weak Accept / Borderline -1: Weak Reject -2: Reject -3: Strong Reject (Serious technical flaws or incorrect)

Weak Accept

2. Please summarize the main contributions of this paper and its significance in your own words.

This paper shows that autonomous agentic AI can exploit a structural weakness in enterprise monitoring systems. Traditional digital forensics assumes malicious actions leave detectable traces. However, fast AI agents can execute anti-forensic actions between periodic monitoring scans.

The authors present Anti-Gravity, a proof-of-concept agent that performs timestomping, secure wiping, and log sanitisation in under 300 ms. The paper also confirms that kernel-level auditing can still capture traces.

3. Novelty and Significance (1-5): 1 is the worst, 5 is the best. "How novel and original are the ideas, methods, or results presented?" "How significant are the contributions to the field?"

3

4. Technical Quality and Correctness (1-5): 1 is the worst, 5 is the best "Is the technical content correct and sound?" "Are the claims well-supported by evidence (theoretical proofs, experiments, etc.)?" "Is the methodology appropriate and rigorously applied?"

3

5. Clarity and Presentation (1-5): 1 is the worst, 5 is the best "Is the paper well-written and clearly organized?" "Is it easy to understand the main points?"

4

6. Comments for the Authors

This is a timely study addressing an important emerging threat. This work demonstrates the feasibility of AI driven forensic attacks and the framing via Nyquist Shannon theorem is interesting. To strengthen the work, I would suggest to broaden the experimental scope, explore more SIEM platforms and vary time intervals, to further assess how performance changes as the time gap narrows. Additionally, if can compare the agent's execution time with that of a skilled human performing sasme tasks, to include timing baseline.

8. Your confidence for the review

high